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The traction boundary value problem for the elastostatic semi-infinite strip; existence of solution, and completeness of the Papkovich–Fadle eigenfunctions

R. D. GREGORY
*Department of Mathematics, University of Manchester, England**

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ABSTRACT

The semi-infinite strip $x \geq 0$, $-1 \leq y \leq 1$ is in equilibrium under no body forces, with the sides $y = \pm 1$, $x > 0$ free of tractions, and on the end $x = 0$

$$\sigma_{xx}(0, y) = f(y),$$
$$\sigma_{xy}(0, y) = g(y),$$

where $f(y)$, $g(y)$ are independent, self-equilibrating tractions prescribed for $y \in [-1, 1]$. A rigorous proof is given that if f'', g'' are of bounded variation on $[-1, 1]$, then this traction boundary value problem posesses a solution, and the stress field of this solution may be expanded, *even on* $x = 0$, as a convergent series of the Papkovich–Fadle eigenfunctions. **Thus these eigenfunctions are complete for the expansion of such data** $\{f, g\}$.

1. Introduction

Boundary value problems for the elastostatic semi-infinite strip have received much attention because of their application to the bending of plates, the stretching extension, bending, and flexure of cantilever beams, and the estimation of "boundary layer" effects in plate and shell theories. These practical problems have in turn generated a strong mathematical interest, particularly in the two fundamental problems with non-mixed boundary conditions, which have proved so resistant to analysis.

Since the corresponding boundary value problems for the *infinite* strip are readily solved by Fourier transforms, we may (without loss of generality) suppose that on the sides $y = \pm 1$, $x > 0$ we have homogeneous boundary conditions, which we shall take to be *traction-free.* (Other side conditions may be similarly treated.) The

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resulting end-data problems for the semi-infinite strip may be divided into two classes: mixed (or "easy") end conditions, and non-mixed (or "hard") end conditions. The mixed end conditions in plane elasticity are those in which either

(a) σ_{xx}, v or (b) σ_{xy}, u

are prescribed on $x = 0$; corresponding end conditions in plate bending would be w, M_{xx} (equivalently w, w_{xx}) prescribed on $x = 0$, where w is the normal deflection. These problems are "easy" essentially because of a symmetry inherent in the corresponding Green's function. Smith [18] (who considers the mixed plate bending problem) obtains a solution as a sum of the Papkovich–Fadle eigenfunctions for the strip, and proves that these eigenfunctions are complete for a rather restricted class of data $w_{xx}(0, y)$, $w_{yy}(0, y)$. Recently Joseph [14], and Joseph and Sturges [15] have pointed out that the conditions required by Smith's completeness proof are not usually satisfied in practical applications. They observed however that even if Smith's conditions are not met, then the corresponding eigenfunction series still converge, perhaps to the data. The author (Gregory [9]) has since proved that these series *do* converge to the prescribed data. In the context of plane elasticity, Johnson and Little [13] have used formal eigenfunction series to treat the mixed end-data problems (a), (b) (above). Although they do not give the necessary proof of completeness, there is little doubt that such a proof could be constructed along the lines of [18] or [9].

For the non-mixed end conditions much less is known, despite the large volume of published work.[1] In particular, existence of solution and completeness of the Papkovich–Fadle eigenfunctions for non-mixed end-data has never been proved. Attempts to analyze these problems lead almost invariably to an infinite system of coupled linear algebraic equations (or an integral equation) which does not belong to any recognizable class for which some theory is known.[2] For instance the treatments of the traction end-data problem by Benthem [1], Gaydon and Shepherd [6], and Johnson and Little [13] lead to (non-standard) infinite systems of coupled linear equations for the coefficients of the Papkovich–Fadle eigenfunction series solution; the treatments of the displacement end-data problem by G. D. Gupta [10], and Bogy [3] lead to singular integral equations with "generalized" Cauchy kernels. No general theory exists for such equations (see Erdogan, Gupta, and Cook [5]).

Only in the work of Teodorescu [20] on the traction end-data problem is a system of equations obtained of a standard form. Teodorescu was able to show that his system of equations does possess a unique bounded solution, but his corresponding expressions for the stresses in the strip do not necessarily converge on the strip boundary, so that it does not seem possible to tell whether this constructed solution takes the correct boundary values or not. [Teodorescu's proof of the regularity of this system of equations was also rediscovered independently by S. C. Gupta [11].]

In the present paper, we give the first rigorous proof of existence of solution of the *traction* end-data problem for the semi-infinite strip. This then enables us to

[1] The author will make no attempt to give a complete bibliography. Bogy [3] gives a good collection of references.

[2] Of course the author is not saying that the *numerical* results obtained by these writers are all incorrect. The methods used seem to be numerically successful, but analytical justification is lacking.

establish the long-conjectured result that the Papkovich–Fadle eigenfunctions are *complete* for the expansion of traction end-data.

For the benefit of the reader, we include here a summary of the existence and expansion results obtained; the details appear in the text in Theorems 7, 8 (and corollary), 9, 10 (and corollary), and 11.

In the following theorem, "problem P for the data $\{f, g\}$" means the problem of determining the state of stress in the semi-infinite strip $x \geq 0$, $-1 \leq y \leq 1$, when the sides $y = \pm 1$, $x > 0$ are traction free, and on the end $x = 0$

$$\sigma_{xx}(0, y) = f(y), \qquad (-1 < y < 1), \tag{1.1}$$

$$\sigma_{xy}(0, y) = g(y), \qquad (-1 < y < 1). \tag{1.2}$$

Solely to simplify the writing, we shall assume f is even and g is odd, so that the resulting stress field is symmetrical about $y = 0$. Corresponding results hold for anti-symmetrical data.

THEOREM. (A) *Suppose $f(y)$ is even, $g(y)$ is odd and that $f''(y)$, $g''(y)$ are of bounded variation on $[-1, 1]$. Then problem P possesses a solution for the data $\{f, g\}$.*

(B) *This solution may be expanded as a series of the Papkovich–Fadle eigenfunctions in the form*

$$\chi(\mathbf{r}) = \sum_{\lambda \in L} A_\lambda E_\lambda(y) e^{-(1/2)\lambda x}, \qquad (x \geq 0, -1 \leq y \leq 1) \tag{1.3}$$

where $\chi(\mathbf{r})$ is the Airy stress function, and $E_\lambda(y)$, L are defined in Appendix 1. In particular, the data tractions on $x = 0$ have expansions of the form

$$f(y) = \sum_{\lambda \in L} A_\lambda E_\lambda''(y), \tag{1.4}$$

$$g(y) = \sum_{\lambda \in L} A_\lambda (\tfrac{1}{2}\lambda) E_\lambda'(y), \tag{1.5}$$

$(-1 < y < 1)$. **Thus the eigenfunction pairs $\{\boldsymbol{E}_\lambda''(\mathbf{y}), \tfrac{1}{2}\boldsymbol{\lambda}\boldsymbol{E}_\lambda'(\mathbf{y})\}$ $(\boldsymbol{\lambda} \in \boldsymbol{L})$ are complete for such data $\{\boldsymbol{f}, \boldsymbol{g}\}$.**

(C) *The rapidity of convergence of the series* (1.4), (1.5) *to the data tractions f, g, depends on the smoothness of f, g and the values of g (but not f) at $y = \pm 1$.*

Case (i). If f''', g''' are of bounded variation on $[-1, 1]$ and $g(\pm 1) = 0$, then

$$A_{\lambda_n} = O\left(\frac{1}{n^{\beta+2}}\right) \tag{1.6}$$

as $n \to \infty$, (where $\beta \doteq 2.74$) and the series (1.4), (1.5) are absolutely-uniformly convergent for $y \in [-1, 1]$, being uniformly majorized by the series

$$\sum \frac{C}{n^{\beta-1} \log n}, \tag{1.7}$$

where C is a constant independent of n.

[Note: $L = \{\lambda_n\} \cup \{\bar{\lambda}_n\}$. See Appendix 1]

Case (ii). If f'', g'' are of bounded variation on $[-1, 1]$ and $g(\pm 1) = 0$, then

$$A_{\lambda_n} = O\left(\frac{1}{n^4 \log n}\right) \tag{1.8}$$

as $n \to \infty$, and the series (1.4), (1.5) are absolutely-uniformly convergent for $y \in [-1, 1]$, being uniformly majorized by the series

$$\sum \frac{C}{n(\log n)^2}, \tag{1.9}$$

where C is a constant independent of n.

Case (iii). If f'', g'' are of bounded variation on $[-1, 1]$ and $g(\pm 1) \neq 0$, then the series (1.4), (1.5) are conditionally convergent[3] for each $y \in (-1, 1)$ and uniformly convergent only for y bounded away from ± 1.

Notes. (a) The rapidity of convergence in Cases (i), (iii) is not in general improved by imposing further *smoothness* restrictions on f, g.

(b) In case (iii) the series (1.4) diverges when $y = \pm 1$, while (1.5) converges to zero, *not* $g(\pm 1)$.

(c) It follows by continuity that in Cases (i), (ii) the series (1.4), (1.5) represent the data tractions f, g on the full *closed* interval $[-1, 1]$.

We conclude this introduction by examining some particular examples discussed by other authors, and predicting the rapidity of convergence of the eigenfunction expansions (1.4), (1.5) to the prescribed data tractions in each case.

(1) $f = -3y + 10y^3 - 7y^5$, $g = -\sin \pi y$ (Johnson and Little [13]). Here $f, g \in C^\infty[-1, 1]$ and $g(\pm 1) = 0$, so we have Case (i) above. (In this example $f(\pm 1) = 0$ also, but this has no significance.)

(2) $f = -\frac{1}{4}\pi^2 \cos(\frac{1}{2}\pi y)$, $g = 0$ (Benthem [1]). These tractions are not self-equilibrating, but they become so if we subtract off the simple tension $\sigma_{xx} = -\frac{1}{2}\pi$; we are then left with Case (i).

(3) $f = 1 - 2|y|$, $g = 0$ (Gaydon and Shepherd [6]). Since $f'(0)$ does not exist, we are not able to make any prediction in this case. The remaining three examples considered by Gaydon and Shepherd are all Case (i).

2. Statement of the problem

Consider the plane shaded region in Figure 1, which depicts a semi-infinite strip of homogeneous, isotropic, linearly elastic material. The strip is supposed to be in equilibrium in a state of generalized plane stress under no body forces, with its sides $y = \pm 1$ free of tractions, and loaded at the end $x = 0$ by prescribed, self-equilibrating tractions. The problem to be solved is to determine the resulting state of stress in the strip, and to show that this stress field may be expanded (even on the end $x = 0$) as a convergent series of the Papkovich–Fadle eigenfunctions for the strip.

In what follows, $\mathscr{D}$ will denote the *open* region $x > 0$, $-1 < y < 1$, $\bar{\mathscr{D}}$ will denote

[3] See Gregory [9], Section 4, for the order in which the terms are to be taken in this case.

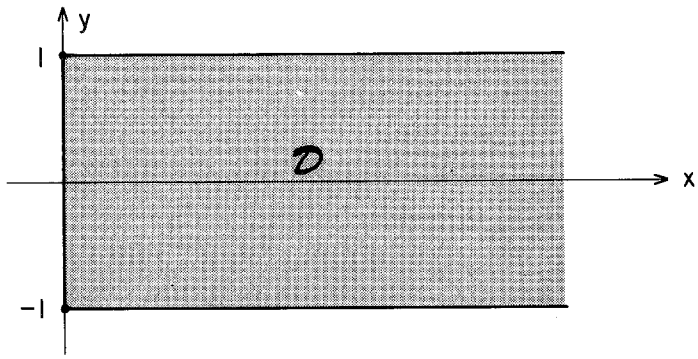


Figure 1. The semi-infinite strip $\mathcal{D}$.

the closure of $\mathcal{D}$, and $\tilde{\mathcal{D}}$ will denote the closure of $\mathcal{D}$ with the two points $(0, \pm 1)$ removed. Where convenient, suffix notation (together with the summation convention) will be used, the range of the suffices being $1 \leq i, j, k \leq 2$.

Statement of the problem P. We seek a two-dimensional, symmetric stress tensor field $\boldsymbol{\sigma}(x, y)$ satisfying the following conditions:

P1. In $\mathcal{D}$, $\boldsymbol{\sigma}$ is twice continuously differentiable and satisfies the equations of equilibrium under no body forces, namely,

$$\frac{\partial \sigma_{ij}}{\partial x_j} = 0. \tag{2.1}$$

P2. $\boldsymbol{\sigma}$ is continuous in $\tilde{\mathcal{D}}$ and satisfies the prescribed conditions of surface traction on the boundary of $\mathcal{D}$, that is

$$\sigma_{xy} = \sigma_{yy} = 0 \quad \text{on} \quad y = \pm 1, \qquad x > 0, \tag{2.2}$$

whilst on the end $x = 0$

$$\sigma_{xx}(0, y) = f(y), \tag{2.3}$$

$$\sigma_{xy}(0, y) = g(y), \tag{2.4}$$

$(-1 < y < 1)$, where $f(y)$, $g(y)$ are "arbitrary" independent functions prescribed on $(-1, 1)$, which we shall call the *data*.

P3. $\boldsymbol{\sigma}$ satisfies the two dimensional condition of compatibility

$$\nabla^2(\sigma_{xx} + \sigma_{yy}) = 0. \tag{2.5}$$

This ensures the existence of a displacement field $\boldsymbol{u}$ related to $\boldsymbol{\sigma}$ by

$$\sigma_{ij} = \left(\frac{2\lambda\mu}{\lambda + 2\mu}\right) e_{kk}\delta_{ij} + 2\mu e_{ij}. \tag{2.6}$$

where

$$e_{ij} = \frac{1}{2}\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right), \tag{2.7}$$

λ, μ being the Lamé moduli of the solid. We will assume that $\mu > 0$, $3\lambda + 2\mu > 0$, thus ensuring that the strain energy density of the solid is positive definite.

P4.

$$W \equiv \iint_{\mathscr{D}} \mathscr{W}\, dx\, dy < \infty, \tag{2.8}$$

where $\mathscr{W}$ denotes the strain energy density of the deformation, which is given by

$$\mathscr{W} = \tfrac{1}{2} e_{ij} \sigma_{ij} \tag{2.9}$$

Notes. (i) The data functions $\{f(y), g(y)\}$ are not of course completely arbitrary; in the first place we shall always require that these tractions be self-equilibrating in the sense that they have no *resultant* force or moment. This implies that

$$\int_{-1}^{1} f(y)\, dy = \int_{-1}^{1} y f(y)\, dy = \int_{-1}^{1} g(y)\, dy = 0. \tag{2.10}$$

Also some conditions of smoothness will be required; indeed the continuity of $\boldsymbol{\sigma}$ in $\tilde{\mathscr{D}}$ (assumed in P2) implies that we must certainly restrict $\{f, g\}$ to be continuous on $(-1, 1)$. Later, in Theorems 7, 9, 11, we give precise conditions on $\{f, g\}$ sufficient to ensure existence of a solution to Problem P.

(ii) Since, from P2, we have already assumed that $\boldsymbol{\sigma}$ is continuous in $\tilde{\mathscr{D}}$, the condition $W < \infty$ in P4 merely serves to place a restriction on the behaviour of $\boldsymbol{\sigma}$ as $x \to \infty$, and to exclude physically unacceptable singularities at the corner points $(0, \pm 1)$. These restrictions are sufficient to ensure *uniqueness* of solution to Problem P. [The fact that uniqueness follows *solely* under the rather weak additional restriction $W < \infty$ is not trivial but will not be pursued further here. The reader is referred to Gregory [8], Theorems 5 (iii), 10, which justify the necessary modifications to the usual Kirchhof energy argument.]

3. Formulation of Problem P as an integral equation of the first kind in the case when $g \equiv 0$

In this section we will reformulate Problem P as a regular Fredholm integral equation of the first kind, in the case when the tangential traction $g \equiv 0$. It would be possible to carry through the existence theory developed in the following sections with $g \not\equiv 0$, but the present approach seems more convenient. The case with $g \not\equiv 0$ is considered in Section 7.

The derivation which follows is formal. Results established with proper rigour are stated as Theorems.

Let us suppose that a solution to Problem P with data $\{f, 0\}$ exists. Then the Airy stress function $\chi(x, y)$ corresponding to this solution satisfies the following conditions:

(i) In $\mathscr{D}$

$$\nabla^2(\nabla^2\chi) = 0. \tag{3.1}$$

(ii) On $y = \pm 1$, $x > 0$

$$\chi = \chi_y = 0. \tag{3.2}$$

(iii) On $x = 0$, $-1 < y < 1$,

$$\chi(0, y) = p(y), \tag{3.3}$$

$$\chi_x(0, y) = 0, \tag{3.4}$$

where

$$p''(y) = f(y), \tag{3.5}$$

and

$$p(\pm 1) = p'(\pm 1) = 0. \tag{3.6}$$

[The relations (2.10) enable us to take the side conditions in the reduced form (3.2), and ensure the existence of a $p(y)$ satisfying (3.5), (3.6).]

Let $G(\boldsymbol{r}', \boldsymbol{r})$ be the fundamental Green's function for the *infinite* strip $-1 \leq y \leq 1$ with traction free sides; this Green's function has been extensively studied by the author [8], and some of its properties are summarized in Appendix 1. Construct the associated Green's function

$$H(\boldsymbol{r}, \boldsymbol{r}') = G(\boldsymbol{r}, \boldsymbol{r}') + G(\tilde{\boldsymbol{r}}, \boldsymbol{r}'), \tag{3.7}$$

where the point $\tilde{\boldsymbol{r}}$ is the reflection of the point $\boldsymbol{r}$ in the line $x = 0$. Then for $\boldsymbol{r} \in A_{X,\delta}$ (see Figure 2) we have the representation formula

$$\chi(\boldsymbol{r}) = \frac{1}{8\pi} \int_{C_{X,\delta}} \left\{ \chi(\boldsymbol{r}') \frac{\partial}{\partial n'} \nabla'^2 H(\boldsymbol{r}, \boldsymbol{r}') - \frac{\partial}{\partial n'} \chi(\boldsymbol{r}') \nabla'^2 H(\boldsymbol{r}, \boldsymbol{r}') \right.$$
$$\left. + \nabla'^2 \chi(\boldsymbol{r}') \frac{\partial}{\partial n'} H(\boldsymbol{r}, \boldsymbol{r}') - \left(\frac{\partial}{\partial n'} \nabla'^2 \chi(\boldsymbol{r}') \right) H(\boldsymbol{r}, \boldsymbol{r}') \right\} ds', \tag{3.8}$$

which follows by an application of the divergence theorem to the integral on the right. Now the contribution to (3.8) from the sides $y = \pm 1$, $\delta < x < X$ is zero, and we expect that the contributions from the end $x = X$, and the quarter-circles near $(0, \pm 1)$, tend to zero as $X \to \infty$ and $\delta \to 0$ respectively. Also $H(\boldsymbol{r}, \boldsymbol{r}')$ is an even function of x' so that

$$\frac{\partial}{\partial x'} H(\boldsymbol{r}, \boldsymbol{r}') \Big|_{x'=0} = \frac{\partial}{\partial x'} \nabla'^2 H(\boldsymbol{r}, \boldsymbol{r}') \Big|_{x'=0} = 0. \tag{3.9}$$

Finally by using (3.4) and the symmetry of $G(\boldsymbol{r}', \boldsymbol{r})$ we obtain the representation

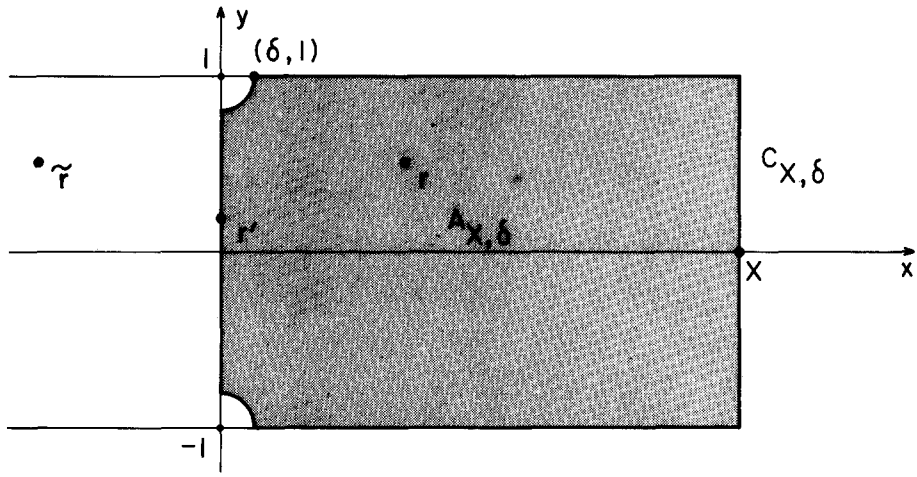


Figure 2. The domain $A_{X,\delta}$ with boundary $C_{X,\delta}$.

$$\chi(\boldsymbol{r}) = \int_{-1}^{1} U(y')G(\boldsymbol{r}', \boldsymbol{r})\Big|_{x'=0} dy', \tag{3.10}$$

for $\boldsymbol{r} \in \mathscr{D}$, where

$$U(y) = \frac{1}{4\pi}\frac{\partial}{\partial x}\nabla^2\chi\Big|_{x=0}. \tag{3.11}$$

By letting $x \to 0+$ in (3.10), we obtain an integral equation satisfied by the unknown function $U(y)$ namely

$$p(y) = \int_{-1}^{1} U(y')G(y', y)\, dy', \tag{3.12}$$

where $p(y)$ is determined from the data $\{f, 0\}$ by (3.5), (3.6), and

$$G(y', y) \equiv G(\boldsymbol{r}', \boldsymbol{r})\Big|_{x'=x=0}. \tag{3.13}$$

(3.12) is the required integral equation for $U(y)$. If a $U(y)$ satisfying (3.12) exists, we may hope that the representation (3.10) then generates the solution to Problem P with data $\{f, 0\}$.

Remarks on the integral equation (3.12)

(3.12) is a *regular* Fredholm integral equation of the first kind for $U(y)$; indeed the kernel $G(y', y)$ is continuous on $[-1, 1]\times[-1, 1]$. Now since it is known (Smithies [19], p. 166) that (3.12) *cannot* possess an $\mathscr{L}^2$-solution for every given $\mathscr{L}^2$-function $p(y)$, this equation may not seem to be a useful starting point for an existence

theory. But this is not the point here; we are not interested in general $\mathscr{L}^2$-data $p(y)$, but in sufficiently differentiable $p(y)$, and this is quite a different matter.

(3.12) is related to a certain class of *singular* integral equations. If (3.10) is differentiated three times in y and then $x \to 0+$, we obtain a (principal value) singular integral equation with a Cauchy type kernel. This equation is of a type whose approximate numerical solution has been treated by Erdogan, Gupta, and Cook [5], and described by them as having a "generalized" Cauchy kernel. There is no general method for the regularization of such equations; in particular the theory developed by Muskelishvili [17], Chap. 14 does not apply. Particular cases have been solved by Bueckner [4], and Biermann [2], but in this work the special singularitity (which characterizes the equation) occurs only at *one* end of the range of integration; the integral equations arising from elastic strip problems possess such singularities at *both* ends of the range, and the author is aware of no case in which such an equation has been regularized.

In Section 4, we will perform such a regularization for (3.12). First however we will prove two results directly.

THEOREM 1. *For each given* $p(y)$, (3.12) *has at most one solution* $U(y) \in C[-1, 1]$.

Proof. Suppose there exists a $U(y) \in C[-1, 1]$ such that

$$\int_{-1}^{1} U(y')G(y', y)\, dy' = 0, \tag{3.14}$$

for $y \in (-1, 1)$. Then it clearly suffices to show that this $U(y) = 0$, $y \in [-1, 1]$. Let $\chi(\boldsymbol{r})$ be *defined* by

$$\chi(\boldsymbol{r}) = \int_{-1}^{1} U(y')G(\boldsymbol{r}', \boldsymbol{r})\Big|_{x'=0} dy', \tag{3.15}$$

where U satisfies (3.14). Then this $\chi(\boldsymbol{r})$ has the following properties:

(i) In $x > 0$, χ is infinitely differentiable and satisfies

$$\nabla^2(\nabla^2\chi) = 0. \tag{3.16}$$

(ii) On $y = \pm 1$, $x > 0$

$$\chi = \chi_y = 0. \tag{3.17}$$

(iii) χ and all its derivatives are exponentially small as $x \to \infty$, uniformly for $-1 \le y \le 1$. (3.18)

(iv) χ, χ_x are continuous in $x \ge 0$, $-1 < y < 1$ and

$$\chi(0, y) = \chi_x(0, y) = 0, \qquad (-1 < y < 1). \tag{3.19}$$

(v) $\chi \to 0$ as $\boldsymbol{r} \to (0, \pm 1)$ through $\bar{\mathscr{D}}$. (3.20)

(i), (ii), (iii) follow most easily from the bi-linear expansion (A1.4) of $G(\boldsymbol{r}', \boldsymbol{r})$ given in Appendix 1. In (iv), (v) we have the required continuity since (from Gregory [8], Theorem 1) $G(\boldsymbol{r}', \boldsymbol{r})$, $G_x(\boldsymbol{r}', \boldsymbol{r})$ are continuous in $\bar{\mathscr{D}} \times \bar{\mathscr{D}}$. (3.19), (3.20) then follow by

noting that

$$G(\boldsymbol{r}', \boldsymbol{r})\Big|_{\boldsymbol{r}=(0,\pm 1)} = G_x(\boldsymbol{r}', \boldsymbol{r})\Big|_{x'=x=0} = 0, \tag{3.21}$$

and by using (3.14).

Now consider the identity

$$\int_{C_{X,\delta}} \left\{\frac{\partial \chi}{\partial n}\nabla^2\chi - \chi\frac{\partial}{\partial n}\nabla^2\chi\right\} ds = \int_{A_{X,\delta}} (\nabla^2\chi)^2\, dx\, dy, \tag{3.22}$$

where $A_{X,\delta}$, $C_{X,\delta}$ are shown in Figure 2. [The higher derivatives in (3.22) present no difficulty at $x=0$, since from (3.16), (3.19) χ has a continuation across $x=0$, $-1<y<1$.] Now the contribution to the left hand side of (3.22) from the sides $y=\pm 1$, $\delta<x<X$, and from the end $x=0$, $|y|<1-\delta$ is zero by (3.17), (3.19) respectively. Also (3.18) implies that the contribution from the end $x=X$, $|y|<1$ tends to zero as $X\to\infty$. It seems difficult to estimate the contribution from the quarter-circles directly from (3.15). However (3.16), (3.17), (3.19), (3.20) imply that for $\boldsymbol{r}$ near (0, 1) say, $\chi(\boldsymbol{r})$ satisfies the conditions of Theorem 9 proved in Gregory [8], with the wedge angle $2\alpha = \pi/2$. It follows that

$$\chi(\boldsymbol{r}) = O(d^{1+\beta}) \tag{3.23}$$

as $d(=|\boldsymbol{r}-(0,1)|\to 0$, where $\beta(\doteqdot 2.74)$ is the real part of the "lowest" root of the secular equation $\sin^2\frac{1}{2}\pi\rho = \rho^2$. The formula (3.23) may be differentiated any number of times. The contribution from the quarter-circles is now easily seen to be $O(\delta^{2\beta})\to 0$ as $\delta\to 0$. Hence

$$\iint_{A_{\infty,0}} (\nabla^2\chi)^2\, dx\, dy = 0, \tag{3.24}$$

and by noting (3.17) it follows that

$$\chi(\boldsymbol{r}) = 0 \qquad (\boldsymbol{r}\in\bar{\mathscr{D}}). \tag{3.25}$$

In particular then it follows that for $\boldsymbol{r}\in\mathscr{D}$

$$\frac{\partial}{\partial x}\nabla^2\chi \equiv \int_{-1}^{1} U(y')\frac{\partial}{\partial x}\nabla^2 G(\boldsymbol{r}', \boldsymbol{r})\Big|_{x'=0} dy' = 0. \tag{3.25}$$

Now let $x\to 0+$ in (3.25) for each fixed $y\in(-1, 1)$. This limiting process is examined in Gregory [9], Theorem 2, where it is shown that the integral tends to $4\pi U(y)$ as $x\to 0+$ whenever y is a point of continuity of U. Hence

$$U(y) = 0 \tag{3.26}$$

for $y\in[-1, 1]$ and this completes the Theorem.

THEOREM 2. *Suppose that the equation*

$$\int_{-1}^{1} U(y')G(y', y)\, dy' = p(y) \tag{3.27}$$

has a solution $U \in C[-1, 1]$. *Then the Airy stress function*

$$\chi(\boldsymbol{r}) = \int_{-1}^{1} U(y')G(\boldsymbol{r}', \boldsymbol{r})\Big|_{x'=0} dy' \tag{3.28}$$

generates the unique solution to Problem P for the data $\{p'', 0\}$.

Proof. As in Theorem 1 it is quickly established that the $\chi(\boldsymbol{r})$ given by (3.28) satisfies (3.16), (3.17), (3.18), (3.20), and on $x = 0$, $-1 < y < 1$,

$$\chi(0, y) = p(y), \tag{3.29}$$

$$\chi_x(0, y) = 0. \tag{3.30}$$

It remains to show that the *stress* field generated by χ satisfies the conditions P2 ((2.3), (2.4)) and P4. For instance for $x > 0$,

$$\sigma_{xx}(\boldsymbol{r}) = \chi_{yy}(\boldsymbol{r}) = \int_{-1}^{1} U(y')G_{yy}(\boldsymbol{r}', \boldsymbol{r})\Big|_{x'=0} dy'$$

$$\to \int_{-1}^{1} U(y')G_{yy}(y', y_0)\, dy' = p''(y_0)$$

as $(x, y) \to (0+, y_0)$, $y_0 \in (-1, 1)$, by dominated convergence. Hence $\sigma_{xx} \in C(\tilde{\mathscr{D}})$ and

$$\sigma_{xx}(0, y) = p''(y), \quad y \in (-1, 1) \tag{3.31}$$

Finally $\boldsymbol{\sigma}(\boldsymbol{r})$ is bounded as $\boldsymbol{r} \to (0, \pm 1)$ through $\tilde{\mathscr{D}}$, and this is certainly sufficient to ensure $W < \infty$.

4. Reduction of the equation (3.12) to a Fredholm equation of the second kind with an $\mathscr{L}^2$-kernel

Our method for regularizing (3.12) is based upon the fact that a closely related equation can be inverted *explicitly*.

Consider the plane elastic semi-circular region shown in Figure 3, which is supposed to be in equilibrium under the following boundary tractions:

(i) On $x^2 + y^2 = 1$, $x > 0$,

$$\boldsymbol{\sigma} \cdot \boldsymbol{n} = \boldsymbol{0}. \tag{4.1}$$

(ii) On $x = 0$, $-1 < y < 1$,

$$\sigma_{xx}(0, y) = f(y), \tag{4.2}$$

$$\sigma_{xy}(0, y) = 0, \tag{4.3}$$

where $f(y)$ is an "arbitrary" function prescribed on $(-1, 1)$, satisfying the equilibrium conditions (2.10).

The problem of determining the resulting state of stress in $\mathscr{D}^C$, for the above data $\{f, 0\}$, can be formulated in an analogous way to that developed for the Problem P in Section 3. The only difference is that the "infinite strip" Green's function $G(\boldsymbol{r}', \boldsymbol{r})$

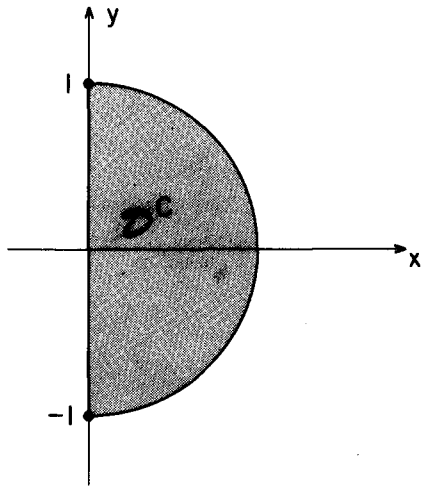


Figure 3. The semi-circular domain $\mathscr{D}^C$ used to generate the "related" integral equation.

must be replaced by the "circle" Green's function $G^C(\boldsymbol{r}', \boldsymbol{r})$. [This "circle" Green's function has a simple form and is given in Love [16], p. 491].

In particular it follows that if the equation

$$\int_{-1}^{1} U(y')G^C(y', y)\,dy' = p(y) \tag{4.4}$$

has a solution $U \in C[-1, 1]$, then the Airy stress function

$$\chi(\boldsymbol{r}) = \int_{-1}^{1} U(y')G^C(\boldsymbol{r}', \boldsymbol{r})\Big|_{x'=0} dy' \tag{4.5}$$

generates the stress field in $\mathscr{D}^C$ corresponding to the data $\{p'', 0\}$. In (4.4) we have used the notation

$$G^C(y', y) \equiv G^C(\boldsymbol{r}', \boldsymbol{r})|_{x'=x=0}$$
$$= (y-y')^2 \log\left(\frac{|y-y'|}{1-yy'}\right) + \frac{1}{2}(1-y^2)(1-y'^2). \tag{4.6}$$

But the above problem for $\mathscr{D}^C$ can be solved in *closed form* and this enables us to solve the integral equation (4.4) explicitly for "arbitrary" $p(y)$. This closed form solution (Jeffery [12]) is derived in a system of bi-polar co-ordinates (ξ, η), where $\zeta(=\xi+i\eta)$ is related to $z(=x+iy)$ by

$$z = i \coth \tfrac{1}{2}\zeta. \tag{4.7}$$

(4.7) maps the strip $\frac{1}{2}\pi < \eta < \pi$, $-\infty < \xi < \infty$ $(1-1)$ onto the domain $\mathscr{D}^C$. The required solution is given in Appendix 2 and has a fairly complicated form involving a Fourier transform in ξ. For this reason it is convenient to consider the "ξ-transform" of the equation (4.4).

If we make the change of variables

$$y = \tanh \tfrac{1}{2}\xi, \tag{4.8}$$

$$y' = \tanh \tfrac{1}{2}\xi', \tag{4.9}$$

then (4.4) becomes

$$\int_{-\infty}^{\infty} \psi(\xi') \frac{\mathcal{G}^C(\xi', \xi)\, d\xi'}{J(\xi')J(\xi)} = \phi(\xi), \tag{4.10}$$

where

$$J(\xi) = (\cosh \xi + 1)^{-1}, \tag{4.11}$$

$$\phi(\xi) = \frac{1}{J(\xi)}\, p\,(\tanh \tfrac{1}{2}\xi), \tag{4.12}$$

$$\psi(\xi) = \{J(\xi)\}^2 U(\tanh \tfrac{1}{2}\xi), \tag{4.13}$$

$$\mathcal{G}^C(\xi', \xi) = G^C(\tanh \tfrac{1}{2}\xi', \tanh \tfrac{1}{2}\xi). \tag{4.14}$$

(4.10) is an equation for the unknown function $\psi(\xi')$, $(-\infty < \xi' < \infty)$, and $\phi(\xi)$ is just the prescribed Jeffery potential on $\eta = \pi$. If we now write

$$\Phi(\alpha) = \int_{-\infty}^{\infty} \phi(\xi) e^{i\alpha\xi}\, d\xi, \tag{4.15}$$

$$\Psi(\alpha) = \int_{-\infty}^{\infty} \psi(\xi) e^{i\alpha\xi}\, d\xi, \tag{4.16}$$

then from Appendix 2,

$$\Psi(\alpha) = M(\alpha)\Phi(\alpha), \tag{4.17}$$

where

$$M(\alpha) = \frac{1}{2\pi} \frac{\alpha(\alpha^2+1) \sinh \tfrac{1}{2}\pi\alpha \cosh \tfrac{1}{2}\pi\alpha}{\sinh^2 \tfrac{1}{2}\pi\alpha - \alpha^2} \tag{4.18}$$

The required solution to equation (4.10) is thus given by

$$\psi(\xi) = \frac{1}{2\pi} \int_{-\infty}^{\infty} M(\alpha)\Phi(\alpha) e^{-i\alpha\xi}\, d\alpha, \tag{4.19}$$

where $M(\alpha)$, $\Phi(\alpha)$ are defined by (4.18), (4.15).

Return now to the integral equation (3.12) and make the changes of variable (4.8)–(4.13). This gives

$$\int_{-\infty}^{\infty} \psi(\xi') \frac{\mathcal{G}(\xi', \xi)\, d\xi'}{J(\xi')J(\xi)} = \phi(\xi), \tag{4.20}$$

where

$$\mathcal{G}(\xi', \xi) = G(\tanh \tfrac{1}{2}\xi', \tanh \tfrac{1}{2}\xi). \tag{4.21}$$

If we now write (4.20) in the form

$$\int_{-\infty}^{\infty} \psi(\xi') \frac{\mathcal{G}^C(\xi', \xi)}{J(\xi')J(\xi)}\, d\xi' = -\int_{-\infty}^{\infty} \psi(\xi') \frac{\mathcal{G}(\xi', \xi) - \mathcal{G}^C(\xi', \xi)}{J(\xi')J(\xi)}\, d\xi' + \phi(\xi), \tag{4.22}$$

then by making use of the known solution (4.19) of the integral equation (4.10), we see that $\Psi(\alpha)$, (defined by (4.16)), is given by

$$\Psi(\alpha) = M(\alpha)\Phi(\alpha) - M(\alpha) \int_{-\infty}^{\infty} \left\{ \int_{-\infty}^{\infty} \psi(\xi') \frac{\mathcal{G}(\xi', \xi) - \mathcal{G}^C(\xi', \xi)}{J(\xi')J(\xi)}\, d\xi' \right\} e^{i\alpha\xi}\, d\xi \tag{4.23}$$

An application of Parseval's theorem then yields

$$\Psi(\alpha) = -\frac{M(\alpha)}{2\pi} \int_{-\infty}^{\infty} H(\alpha', \alpha)\Psi(\alpha')\, d\alpha' + M(\alpha)\Phi(\alpha), \tag{4.24}$$

where

$$H(\alpha', \alpha) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{\mathcal{G}(\xi', \xi) - \mathcal{G}^C(\xi', \xi)}{J(\xi')J(\xi)} e^{i\alpha'\xi'} e^{i\alpha\xi}\, d\xi'\, d\xi. \tag{4.25}$$

(4.24) has the form

$$\Psi(\alpha) = \int_{-\infty}^{\infty} K(\alpha', \alpha)\Psi(\alpha')\, d\alpha' + M(\alpha)\Phi(\alpha), \tag{4.26}$$

where the kernel $K(\alpha', \alpha)$ is given by

$$K(\alpha', \alpha) = -\frac{1}{2\pi} M(\alpha)H(\alpha', \alpha), \tag{4.27}$$

where $H(\alpha', \alpha)$ is defined by (4.25), $M(\alpha)$ by (4.18) and $\Phi(\alpha)$ is determined (via (4.15), (4.12), (3.5)) from the data $\{f, 0\}$ in problem P.

(4.26) is the required Fredholm integral equation of the second kind for the unknown function $\Psi(\alpha)$, $(-\infty < \alpha < \infty)$. If (4.26) has a solution, then we may hope to construct the solution to problem P for the data $\{f, 0\}$ via (4.16), (4.13), (3.28).

Remarks. (i) Although the range of the integral equation (4.26) is the whole real line, this is *not* a singular equation. We will in fact show that $K(\alpha', \alpha) \in \mathcal{L}^2(-\infty, \infty) \times (-\infty, \infty)$ so that the classical $\mathcal{L}^2$-theory applies. (See the remarks in Smithies [19], p. 6).

(ii) In applying Parseval's theorem to obtain (4.24) we have assumed that the data function $f(y)$ is even in y, thus making $\psi(\xi)$ even in ξ. This restriction is inessential, but it does simplify the subsequent writing.

THEOREM 3. *$G(y', y) - G^C(y', y)$ is an analytic function of the two complex variables (y', y) in $L_\gamma \times L_\gamma$ where L_γ is the open lune shown in Figure 3, $(0 < \gamma < \frac{1}{2}\pi)$. Furthermore*

$$G(y', y) - G^C(y', y) = \frac{(1 - y'^2)^2(1 - y^2)^2}{(2 - y' - y)(2 + y' + y)} + E(y', y), \tag{4.28}$$

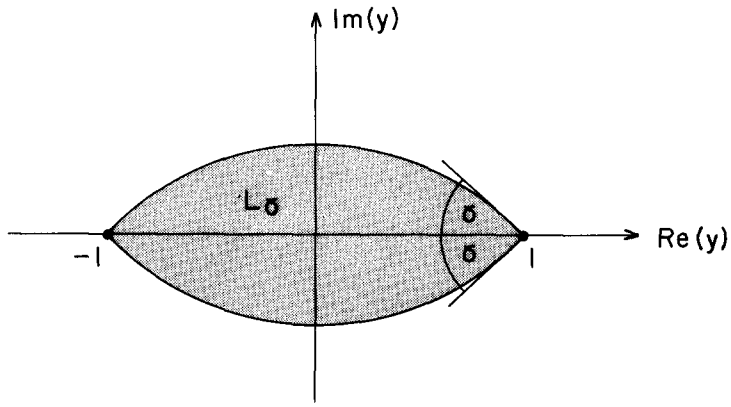


Figure 4. The open lune L_γ in the complex y-plane.

where

$$|E(y', y)| \leq M\,|1-y'^2|^2\,|1-y^2|^2, \qquad (y', y) \in L_\gamma \times L_\gamma, \tag{4.29}$$

and M is independent of y', y.

Outline of proof. It follows from the form of $G(y', y)$ (given in Gregory [8], equation (2.18)) and from (4.6) that

$$G(y', y) - G^C(y', y) = (y-y')^2 \log\left(\frac{1-yy'}{2-y-y'}\right) + \frac{1}{2}(1-y)(1-y')[4-(1+y)(1+y')] + \int_0^\infty (I_1+I_2+I_4)\,dk, \tag{4.30}$$

where the functions I_1, I_2, I_4 (y', y, k) are given in [8]. The integral term is analytic in

$$-1 < \mathrm{Re}\,(y'),\ \mathrm{Re}\,(y) < 1, \tag{4.31}$$

and the log term is analytic in

$$|y'|, |y| < 1, \tag{4.32}$$

so that $G(y', y) - G^C(y', y)$ is in fact analytic in $L_{\pi/2} \times L_{\pi/2}$. The bound (4.28), (4.29) is obtained straightforwardly from (4.30), but it is necessary to restrict $\gamma < \frac{1}{2}\pi$.

COROLLARY

$$|G(y', y) - G^C(y', y)| < M\,|1-y'^2|^{3/2}\,|1-y^2|^{3/2} \tag{4.33}$$

for $(y', y) \in L_\gamma \times L_\gamma$, *where M is independent of* y', y.

Notation. (i) In the plane of complex u (say), the open strip

$$-c < \mathrm{Im}\,(u) < c \tag{4.34}$$

will be denoted by S_c.

(ii) $S_{<c}$ will mean any strip

$$-c+\delta<\mathrm{Im}\,(u)<c-\delta, \tag{4.35}$$

with $\delta>0$.

THEOREM 4. *$H(\alpha', \alpha)$, defined by* (4.25), *is an analytic function of the* two complex *variables α', α in $S_{1/2}\times S_{1/2}$ and*

$$|H(\alpha', \alpha)|<Me^{-\gamma(|\alpha'|+|\alpha|)} \tag{4.36}$$

for $(\alpha', \alpha)\in S_{<1/2}\times S_{<1/2}$, where $0<\gamma<\frac{1}{2}\pi$, and M is independent of α', α.

Proof. It follows from Theorem 3 and the definitions (4.14), (4.21) that $\{\mathscr{G}(\xi', \xi)-\mathscr{G}^C(\xi', \xi)\}/J(\xi')J(\xi)$ is analytic for $(\xi', \xi)\in S_{\pi/2}\times S_{\pi/2}$ and that

$$\left|\frac{\mathscr{G}(\xi', \xi)-\mathscr{G}^C(\xi', \xi)}{J(\xi')J(\xi)}\right|<Me^{-1/2(|\xi'|+|\xi|)} \tag{4.37}$$

for $(\xi', \xi)\in S_{<\pi/2}\times S_{<\pi/2}$. The analyticity of $H(\alpha', \alpha)$ now follows from (4.37), and the bound (4.36) follows by deforming the contours of integration in (4.25) in the complex ξ', ξ-planes.

COROLLARY. *The kernel $K(\alpha', \alpha)$ satisfies the same conditions as $H(\alpha', \alpha)$ stated in Theorem* 4. *In particular $K(\alpha', \alpha)\in\mathscr{L}^2(-\infty, \infty)\times(-\infty, \infty)$.*

Remark. It is tempting in the subsequent analysis to express $H(\alpha', \alpha)$ as the difference of the double Fourier transforms of $\mathscr{G}(\xi', \xi)/J(\xi')J(\xi)$ and $\mathscr{G}^C(\xi', \xi)/J(\xi')J(\xi)$. However these separate transforms *do not exist.*

5. The equation (4.26) has an $\mathscr{L}^2$-solution

Since we now know that the kernel $K(\alpha', \alpha)\in\mathscr{L}^2$, it suffices to consider the corresponding homogeneous equation.

THEOREM 5. *The only L^2-solution of the equation*

$$\Psi(\alpha)=\int_{-\infty}^{\infty}K(\alpha', \alpha)\Psi(\alpha')\,d\alpha' \tag{5.1}$$

is the trivial one $\Psi(\alpha)\equiv 0$.

Proof. Let $\Psi(\alpha)$ be an $\mathscr{L}^2$-solution of (5.1). Then by Theorem 4 (Corollary), it follows from (5.1) that $\Psi(\alpha)$ is analytic in $S_{1/2}$ and

$$\Psi(\alpha)=O(e^{-\gamma|\alpha|}) \tag{5.2}$$

as $|\alpha|\to\infty$ in $S_{<1/2}$, for each γ, $0<\gamma<\frac{1}{2}\pi$.

IMPORTANT REMARK. On the information (5.2) *alone* one can only construct a $U(y)$ (via (4.16), (4.13)) which *may* have singularities at $y=\pm 1$ like $(1-y^2)^{-3/2}$, and this certainly cannot be used in Theorem 2 to construct the corresponding stress

field. An analogous difficulty appeared in Teodorescu's work [20], where although a solution of his system of linear equations was established, it was not possible to complete his theory by constructing the corresponding stress field.

In the present case this difficulty *can* be overcome, but only with some effort; the necessary result is established in Appendix 3, Theorem 12. It follows from (5.2) and Theorem 12 that

$$R(\alpha) \equiv \int_{-\infty}^{\infty} H(\alpha', \alpha)\Psi(\alpha')\, d\alpha' \tag{5.3}$$

is analytic in $S_{3/2}$, except possibly for simple poles at $\pm i$, and that

$$R(\alpha) = O(e^{-\gamma|\alpha|}) \tag{5.4}$$

as $|\alpha| \to \infty$ in $S_{<3/2}$, for each γ, $0 < \gamma < \frac{1}{2}\pi$.

Hence since

$$K(\alpha', \alpha) = -\frac{1}{2\pi} M(\alpha)H(\alpha', \alpha), \tag{5.5}$$

($M(\alpha)$ being given by (4.18)), it follows from (5.1), (5.3) that $\Psi(\alpha)$ must be analytic in $S_{3/2}$ and

$$\Psi(\alpha) = O(e^{-\gamma|\alpha|}) \tag{5.6}$$

as $|\alpha| \to \infty$ in $S_{<3/2}$. [This follows since the poles of $R(\alpha)$ are annihilated by the corresponding zeros of $M(\alpha)$]. This process of "continuation by iteration" could proceed indefinitely except for the fact that $M(\alpha)$ itself has simple poles at the (complex) zeros of

$$\sinh^2 \tfrac{1}{2}\pi\alpha = \alpha^2. \tag{5.7}$$

These complex zeros are discussed in [7], Appendix 1; $\pm\alpha_1$, $\pm\bar{\alpha}_1$, the zeros closest to the real axis of α, have imaginary part with magnitude $\beta = 2.74\ldots$. Hence by two further applications of the iteration process above, we deduce that $\Psi(\alpha)$ is analytic in S_3 (say), except for simple poles at $\pm\alpha_1$, $\pm\bar{\alpha}_1$, and

$$\Psi(\alpha) = O(e^{-\gamma|\alpha|}) \tag{5.8}$$

as $|\alpha| \to \infty$ in S_3.

Now *define* $\Phi^C(\alpha)$ by

$$\Phi^C(\alpha) = \frac{\Psi(\alpha)}{M(\alpha)}. \tag{5.9}$$

Then this $\Phi^C(\alpha)$ is analytic in S_3, except for simple poles at $\pm i$, $\pm 2i$, $\pm 3i$, and

$$\Phi^C(\alpha) = O(e^{-\gamma|\alpha|}) \tag{5.10}$$

as $|\alpha| \to \infty$ in S_3.

Now substitute this $\Phi^C(\alpha)$ into the expression (A2.5) in Appendix 2. This then generates the Jeffery potential $\phi^C(\xi, \eta)$ of a certain stress field in $\mathscr{D}^C$ (see Figure 3).

From the above properties of $\Phi^C(\alpha)$ it quickly follows that this stress field satisfies (4.1), (4.3), and that $\phi^C(\xi)$ $(=\phi^C(\xi, \pi))$ and $\psi(\xi)$ (defined by (A2.6)) satisfy the integral equation (4.10), namely

$$\int_{-\infty}^{\infty} \psi(\xi') \frac{\mathscr{G}^C(\xi', \xi)}{J(\xi')J(\xi)}\, d\xi' = \phi^C(\xi). \tag{5.11}$$

If we now divide (5.1) through by $M(\alpha)$, use (A2.8), and take the inverse Fourier transform, we obtain

$$\phi^C(\xi) = -\int_{-\infty}^{\infty} \psi(\xi') \frac{\mathscr{G}(\xi', \xi) - \mathscr{G}^C(\xi', \xi)}{J(\xi')J(\xi)}\, d\xi'. \tag{5.12}$$

(5.11), (5.12) imply that

$$\int_{-\infty}^{\infty} \psi(\xi') \frac{\mathscr{G}(\xi', \xi)}{J(\xi')J(\xi)}\, d\xi' = 0 \tag{5.13}$$

i.e.

$$\int_{-1}^{1} U(y')G(y', y)\, dy' = 0, \tag{5.14}$$

where $U(y)$ is related to $\psi(\xi)$ by (4.13). Moreover the known regularity of $\Psi(\alpha)$ implies that $U(y) \in C[-1, 1]$ and hence by Theorem 1

$$U(y) \equiv 0. \tag{5.15}$$

This implies that $\psi(\xi) \equiv 0$ and hence that $\Psi(\alpha) \equiv 0$, and this completes the theorem.

THEOREM 6. *The equation*

$$\Psi(\alpha) = \int_{-\infty}^{\infty} K(\alpha', \alpha)\Psi(\alpha')\, d\alpha' + M(\alpha)\Phi(\alpha) \tag{5.16}$$

has an $\mathscr{L}^2$-solution $\Psi(\alpha)$ for any $\mathscr{L}^2$-function $\Phi(\alpha)$ which satisfies

$$\Phi(\alpha) = O\left(\frac{1}{\alpha^4}\right), \tag{5.17}$$

as $|\alpha| \to \infty$.

Proof. If $\Phi(\alpha) \in \mathscr{L}^2(-\infty, \infty)$ and satisfies (5.17), then $M(\alpha)\Phi(\alpha) \in \mathscr{L}^2(-\infty, \infty)$. Theorem 5 and the Fredholm alternative then imply that (5.16) has an $\mathscr{L}^2$-solution $\Psi(\alpha)$.

6. To construct the solution to problem P for the data $\{f, 0\}$, and the expansion of this solution as a series of Papkovich–Fadle eigenfunctions

THEOREM 7. *Let $f''(y) \in BV[-1, 1]$.*[4] *Then Problem P possesses a solution for the data $\{f, 0\}$.*

[4] This notation means that f'' is of bounded variation on $[-1, 1]$. In particular then $f \in C^1[-1, 1]$.

Proof. If $f'' \in BV[-1, 1]$, then $p(y)$, defined by (3.5), (3.6), is such that

$$p'''' \in BV[-1, 1], \tag{6.1}$$

with

$$p(\pm 1) = p'(\pm 1) = 0. \tag{6.2}$$

(6.1), (6.2) imply that $\Phi(\alpha)$, defined by (4.15), (4.12), is analytic in S_3, except for simple poles at $\pm i$, $\pm 2i$, and

$$\Phi(\alpha) = O\left(\frac{1}{\alpha^5}\right) \tag{6.3}$$

as $|\alpha| \to \infty$ in $S_{<3}$.

Hence by Theorem 6, the integral equation (5.16) has an $\mathscr{L}^2$-solution $\Psi(\alpha)$. By the same continuation procedure employed in Theorem 5, it follows that this $\Psi(\alpha)$ is analytic in S_3, except for simple poles at $\pm\alpha_1$, $\pm\bar{\alpha}_1$, and

$$\Psi(\alpha) = O\left(\frac{1}{\alpha_2}\right) \tag{6.4}$$

as $|\alpha| \to \infty$ in $S_{<3}$.

If we define $\Phi^C(\alpha)$ by (5.9) and follow the same procedure employed in Theorem 5, we obtain the two equations

$$\int_{-\infty}^{\infty} \psi(\xi') \frac{\mathscr{G}^C(\xi', \xi)}{J(\xi')J(\xi)}\, d\xi' = \phi^C(\xi), \tag{6.5}$$

$$\phi^C(\xi) = -\int_{-\infty}^{\infty} \psi(\xi') \frac{\mathscr{G}(\xi', \xi) - \mathscr{G}^C(\xi', \xi)}{J(\xi')J(\xi)}\, d\xi' + \phi(\xi). \tag{6.6}$$

It follows from (6.5), (6.6) that

$$\int_{-\infty}^{\infty} \psi(\xi') \frac{\mathscr{G}(\xi', \xi)}{J(\xi')J(\xi)}\, d\xi' = \phi(\xi), \tag{6.7}$$

and so

$$\int_{-1}^{1} U(y')G(y', y)\, dy' = p(y), \tag{6.8}$$

where

$$U(\tanh \tfrac{1}{2}\xi) = \psi(\xi)\{J(\xi)\}^{-2}. \tag{6.9}$$

Now it follows from the regularity and bound (6.4) on $\Psi(\alpha)$ that $U(y) \in C[-1, 1]$ and so by Theorem 2, the solution to problem P for the data $\{f, 0\}$ is generated by the Airy stress function

$$\chi(\boldsymbol{r}) = \int_{-1}^{1} U(y')G(\boldsymbol{r}', \boldsymbol{r})|_{x'=0}\, dy'. \tag{6.10}$$

Expansion of the solution in terms of eigenfunctions

It follows from the author's work in [8], Theorems 4, 5, that *any* stress field satisfying P1–P4 has an eigenfunction expansion in the region $x>0$, $-1\leq y\leq 1$, no matter what the end data may be. The principal interest here then is whether this eigenfunction expansion also represents the stress field *on the end* $x=0$. To simplify the writing we will assume that $f(y)$ is even.

THEOREM 8. *Suppose f is even and $f''\in BV[-1,1]$. Then the solution to Problem P for the data $\{f,0\}$ may be expanded as a series of the Papkovich–Fadle eigenfunctions in the form*

$$\chi(\mathbf{r})=\sum_{\lambda\in L} A_\lambda E_\lambda(y)e^{-1/2\lambda x} \tag{6.11}$$

$(x\geq 0, -1\leq y\leq 1)$, where $E_\lambda(y)$, L are defined in Appendix 1. In particular the data stresses on $x=0$ are given by

$$f(y)=\sum_{\lambda\in L} A_\lambda E_\lambda''(y), \tag{6.12}$$

$$0=\sum_{\lambda\in L} A_\lambda(\tfrac{1}{2}\lambda)E_\lambda'(y), \tag{6.13}$$

where the series (6.12), (6.13) are absolutely-uniformly convergent for $y\in[-1,1]$, being uniformly majorized by the convergent series

$$\sum \frac{C}{n(\log n)^2}, \tag{6.14}$$

$(-1\leq y\leq 1)$, where C is a constant independent of n. Thus the eigenfunction pairs $\{E_\lambda'', \tfrac{1}{2}\lambda E_\lambda'\}$ $(\lambda\in L)$, are complete for such data $\{f,0\}$.

Proof. Since f satisfies the conditions of Theorem 7, we know that problem P has a solution for the data $\{f,0\}$ generated by the Airy stress function

$$\chi(\mathbf{r})=\int_{-1}^{1} U(y')G(\mathbf{r}',\mathbf{r})|_{x'=0}\,dy', \tag{6.15}$$

where $U(y)$ $(\in C[-1,1])$ may be identified as

$$U(y)=\frac{1}{4\pi}\frac{\partial}{\partial x}\nabla^2\chi\Big|_{x=0}. \tag{6.16}$$

If we now substitute the bi-linear eigenfunction expansion (A1.4) for $G(\mathbf{r}',\mathbf{r})$ into (6.15) and integrate term by term, we obtain

$$\chi(\mathbf{r})=\sum_{\lambda\in L} A_\lambda E_\lambda(y)e^{-1/2\lambda x}, \tag{6.17}$$

where

$$A_\lambda = \frac{\tan^2 \frac{1}{2}\lambda}{2\lambda^3} \int_{-1}^{1} E_\lambda(y) \frac{\partial}{\partial x} \nabla^2 \chi \Big|_{x=0} dy. \tag{6.18}$$

[The anti-symmetric terms disappear since f is assumed even, and the term by term integration is justified by the uniform convergence of (A1.4) and continuity of $U(y)$].

The series (6.17) for $\chi(r)$ thus converges on $x = 0$, but we are more interested in the corresponding series for the stresses on $x = 0$, and so we need bounds on $\{A_\lambda\}$. Thus far we know that $(\partial/\partial x)\nabla^2\chi|_{x=0} \in C[-1, 1]$, but this by no means implies bounds on $\{A_\lambda\}$ (via (6.18)) sufficient to justify two differentiations of (6.17); hence a subtler argument is needed.

Let

$$J_\lambda = \int_{-1}^{1} E_\lambda(y) \frac{\partial}{\partial x} \nabla^2 \chi \Big|_{x=0} dy \tag{6.19}$$

$$= \int_{-1}^{1} \phi_\lambda \frac{\partial}{\partial x} \nabla^2 \chi \Big|_{x=0} dy$$

$$= \int_{-1}^{1} \nabla^2 \chi \frac{\partial}{\partial x} \phi_\lambda + \chi \frac{\partial}{\partial x} \nabla^2 \phi_\lambda \Big|_{x=0} dy, \tag{6.20}$$

after applying the reciprocity result in [8], Theorem 3(i).

Hence

$$J_\lambda = -\tfrac{1}{2}\lambda \int_{-1}^{1} E_\lambda(y) \nabla^2 \chi + (2\lambda \cos \tfrac{1}{2}\lambda \cos \tfrac{1}{2}\lambda y)\chi|_{x=0} \, dy \tag{6.21}$$

The virtue of this transformation is that the smoothness of $\nabla^2\chi|_{x=0}$ is closely linked to that of the data function $f(y)$; this is not so of $(\partial/\partial x)\nabla^2\chi|_{x=0}$.

LEMMA. *When $g \equiv 0$ and f is even,*

$$\nabla^2 \chi|_{x=0} = 2f(y) - f(1) - \tfrac{1}{2}(y^2 - 1)f'(1) + P(y), \tag{6.22}$$

where $P(y)$ is a regular function of complex y in the domain shown in Figure 5, and as $y \to \pm 1$ through this domain

$$P(y) = O(y^2 - 1)^{\beta - 1}, \tag{6.23}$$

where $\beta (\doteq 2.74)$ is defined after (5.7).

Proof

Let us compare the solution to Problem P (with data $\{f, 0\}$) with the right-angled wedge solution found in Appendix 4. To do this we identify the corner $(0, 1)$ (say) with $r = 0$, the line $y = 1$, $x \geq 0$ with $\theta = \frac{1}{2}\pi$, and the line $x = 0$, $y \leq 1$ with $\theta = 0$. The data $\{F, G\}$ on $\theta = 0$ is taken to be

$$G \equiv 0, \tag{6.24}$$

$$F(r) = f(1 - r), \qquad (0 \leq r \leq 2), \tag{6.25}$$

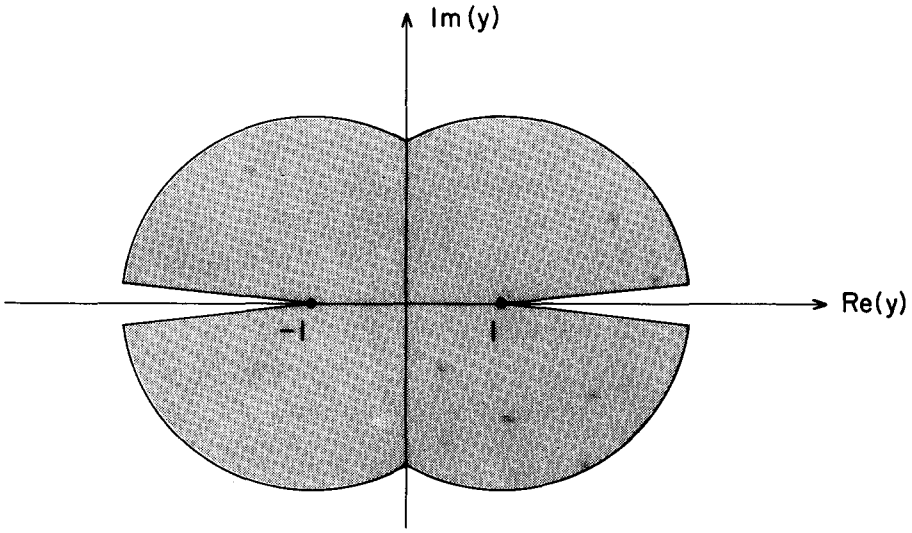


Figure 5. The domain of regularity of $P(y)$, $Q(y)$, $R(y)$.

and $F(r)$ is continued for $r>2$ so that

(i) $F(r)=0$, for all sufficiently large r, (6.26)

(ii) $F''(r)\in BV[0,\infty)$, (6.27)

(iii) $$\int_0^\infty F\,dr=\int_0^\infty rF\,dr=0. \tag{6.28}$$

Then (A4.4) certainly generates a continuous stress field in the wedge $r\geq 0$, $0\leq\theta\leq\frac{1}{2}\pi$, which satisfies (A4.1)–(A4.3). Now for *this* stress field, $\nabla^2\chi|_{\theta=0}$ is given by (A4.6) where the integral term is a regular function of *complex* r in $-\pi<\arg(r)<\pi$, $|r|>0$. It also quickly follows that as $r\to 0$,

$$\nabla^2\chi|_{\theta=0}\to F(0), \tag{6.29}$$

$$\frac{\partial}{\partial x}\nabla^2\chi\Big|_{\theta=0}\to F'(0), \tag{6.30}$$

and so for this wedge solution

$$\nabla^2\chi|_{\theta=0}=2F(r)-F(0)-rF'(0)+P^*(r), \tag{6.31}$$

where $P^*(r)$ is regular in $-\pi<\arg(r)<\pi$, $|r|>0$, and as $|r|\to 0$ with $\arg r$ bounded away from $\pm\pi$,

$$P^*(r)=O(r^{\beta-1}), \tag{6.32}$$

where $\beta(\doteq 2.74)$ is the real part of the "lowest" root of $\sin^2\frac{1}{2}\pi p=p^2$.

But for $r<2$, this wedge solution differs from the problem P solution by a field satisfying *stress free* boundary conditions on $\theta=0,\frac{1}{2}\pi$, and possessing bounded strain energy near $r=0$. This difference field thus satisfies the conditions of Theorem 10

(Gregory [8]) and hence has an expansion in terms of the Williams eigenfunctions for a wedge of angle $\frac{1}{2}\pi$. It follows that for $|r|<2$, (6.31) remains true for the problem P solution with a redefined $P^*(r)$.

The corner $(0,-1)$ may be similarly treated, and this establishes the lemma.

Returning to the proof of Theorem 8, if we substitute (6.22) into (6.21) we obtain (after some integrations by parts)

$$J_\lambda = -\tfrac{1}{2}\lambda \int_{-1}^{1} E_\lambda(y)P(y)\,dy + \frac{16}{\lambda^2}\int_{-1}^{1} \cos\tfrac{1}{2}\lambda \cos\tfrac{1}{2}\lambda y(3f''(y)-2f'(1))\,dy - \frac{16}{\lambda^3}\int_{-1}^{1} E_\lambda''(y)f''(y)\,dy, \tag{6.33}$$

where we have also used the identity

$$E_\lambda''(y)+\tfrac{1}{4}\lambda^2 E_\lambda(y) = 2\lambda \cos\tfrac{1}{2}\lambda \cos\tfrac{1}{2}\lambda y. \tag{6.34}$$

Now let $|\lambda|\to\infty$ through $\{\lambda_n\}$, where $\{\lambda_n\}$ are defined in Appendix 1. The argument used in Gregory [9] Theorem 5 holds good, in which case the second and third terms in (6.33) are $O(1/n^2)$, $O(1/n\log n)$ as $n\to\infty$ respectively. Since we have the regularity and bound (6.23) for $P(y)$, the first term in (6.33) may be estimated by using complex variable techniques followed by Watson's Lemma. The details are straightforward and yield the result

$$\int_{-1}^{1} E_{\lambda_n}(y)P(y)\,dy = O\left(\frac{1}{n^\beta}\right) \tag{6.35}$$

as $n\to\infty$, (where $\beta \doteqdot 2.74$).

On combining these results, we find that

$$J_{\lambda_n} = O\left(\frac{1}{n\log n}\right), \tag{6.36}$$

and hence

$$A_{\lambda_n} = O\left(\frac{1}{n^4\log n}\right) \tag{6.37}$$

as $n\to\infty$.

It now follows that the terms in the series (6.12), (6.13) arising from $\{\lambda_n\}$ are $O(1/n(\log n)^2)$ as $n\to\infty$, uniformly for $y\in[-1,1]$. The same bound holds for terms arising from $\{\bar{\lambda}_n\}$ and this completes the theorem.

COROLLARY. *If* $f'''\in BV[-1,1]$ *then*

$$A_{\lambda_n} = O\left(\frac{1}{n^{\beta+2}}\right) \tag{6.38}$$

as $n \to \infty$, *and the terms of* (6.12), (6.13) *are*

$$O\left(\frac{1}{n^{\beta-1}\log n}\right) \tag{6.39}$$

as $n \to \infty$, *uniformly for* $y \in [-1, 1]$. (*This rate of convergence is not in general improved by placing further smoothness restrictions on* f).

Proof. If $f''' \in BV[-1, 1]$, the last term in (6.33) may be integrated by parts again and gives a contribution to J_{λ_n} of $O(1/n^2 \log n)$ as $n \to \infty$. The largest contribution to J_{λ_n} now arises from the first term.

7. To construct the solution to problem P for the data $\{0, g\}$, and its expansion as a series of Papkovich–Fadle eigenfunctions

The idea of the next theorem is to introduce a special stress field which is known to take the correct value of $g(y) \equiv -\chi_{xy}(0, y)$. This field will however have some residual normal stress on $x = 0$, which we then hope will satisfy the conditions of Theorem 7. However a naive choice for this special field, for example the one which satisfies

$$\chi_{xy} = -g, \tag{7.1}$$

$$\frac{\partial}{\partial x}\nabla^2\chi = 0, \tag{7.2}$$

on $x = 0$, leads to a residual normal stress whose smoothness is not simply related to that of g, and which may even be unbounded near $y = \pm 1$. Hence a more subtle choice is needed. (Solely to simplify the writing, we will assume $g(y)$ is *odd*).

THEOREM 9. *Suppose* $g(y)$ *is odd with* $g(\pm 1) = 0$[5] *and* $g''(y) \in BV[-1, 1]$. *Then problem P possesses a solution for the data* $\{0, g\}$.

Proof. Consider the function $\hat{\chi}(x, y)$, bi-harmonic in the semi-infinite strip $\mathcal{D}$, which is chosen to satisfy the conditions

$$\hat{\chi} = \hat{\chi}_y = 0 \quad \text{on} \quad y = \pm 1, \qquad x > 0, \tag{7.3}$$

$$\hat{\chi}_{xy} = -g(y), \tag{7.4}$$

$$\frac{\partial}{\partial x}\nabla^2\hat{\chi} = 2g'(y) - 3g'(1), \tag{7.5}$$

on $x = 0$, $-1 < y < 1$. The fact that such a $\hat{\chi}$ exists follows from the author's work in Gregory [9], Theorem 3. In fact $\hat{\chi}$ may be expressed immediately as the eigenfunction series

$$\hat{\chi}(x, y) = \sum_{\lambda \in L} A_\lambda \left(-\frac{2}{\lambda}\right) E_\lambda(y) e^{-1/2\lambda x}, \tag{7.6}$$

[5] The restriction $g(\pm 1) = 0$ will be removed in Theorem 11.

where

$$A_\lambda = -\frac{\tan^2 \frac{1}{2}\lambda}{4\lambda^2} \int_{-1}^{1} \{(4 \cos \tfrac{1}{2}\lambda \sin \tfrac{1}{2}\lambda y) g(y) + (2g'(y) - 3g'(1)) E_\lambda(y)\, dy\}, \tag{7.7}$$

This $\hat{\chi}$ generates a stress field, continuous in $\bar{\mathcal{D}}$, and satisfying (7.3)–(7.5). The residual normal stress $\hat{\chi}_{yy}(0, y)$ is thus known to be $C[-1, 1]$, but this is by no means sufficient for Theorem 7 to be applicable.

LEMMA

$$\hat{\chi}_{yy}(0, y) = A + B(y^2 - 1) + Q(y), \tag{7.8}$$

where $Q(y)$ is a regular function of complex y in the domain shown in Figure 5, and as $y \to \pm 1$ through this domain

$$Q(y) = O(1 - y^2)^{2-\varepsilon}, \tag{7.9}$$

for each $\varepsilon > 0$, where A, B are constants.

Proof. This can be deduced from the corresponding right-angled wedge solution, as in Theorem 8 (in this case from (A4.7)).

Even (7.8), (7.9) do not imply that $\hat{\chi}_{yy}(0, y)$ satisfies the conditions (on $f(y)$) in Theorem 7; in particular $f''(\pm 1)$ may not exist. However these conditions are not necessary, only sufficient, and it *does* follow from (7.8), (7.9) that $\hat{\Phi}(\alpha)$ (the ξ-transform of the associated Jeffery potential) satisfies the regularity and bound (6.3). This is also sufficient for the conclusion of Theorem 7 to hold, and this completes the theorem.

THEOREM 10. *Suppose $g(y)$ is odd with $g(\pm 1) = 0$ and $g''(y) \in BV[-1, 1]$. Then the solution to Problem P for the data $\{0, g\}$ may be expanded as a series of the Papkovich–Fadle eigenfunctions in the form*

$$\chi(\mathbf{r}) = \sum_{\lambda \in L} A_\lambda E_\lambda(y) e^{-(1/2)\lambda x}, \tag{7.10}$$

$(x \geq 0, -1 \leq y \leq 1)$. In particular the data stresses on $x = 0$ are given by

$$0 = \sum_{\lambda \in L} A_\lambda E_\lambda''(y), \tag{7.11}$$

$$g(y) = \sum_{\lambda \in L} A_\lambda (\tfrac{1}{2}\lambda) E_\lambda'(y), \tag{7.12}$$

where the series (7.11), (7.12) are absolutely-uniformly convergent for $y \in [-1, 1]$, being uniformly majorized by the convergent series

$$\sum \frac{C}{n(\log n)^2}, \tag{7.13}$$

$(-1 \leq y \leq 1)$, *where* C *is a constant independent of* n. *Thus the eigenfunction pairs* $\{E_\lambda'', \frac{1}{2}\lambda E_\lambda\}$ $(\lambda \in L)$ *are complete for such data* $\{0, g\}$.

Proof. We now know that problem P has a solution for such data $\{0, g\}$ and, by following the procedure in Section 2, this solution may be generated by the Airy stress function

$$\chi(\boldsymbol{r}) = \frac{1}{4\pi} \int_{-1}^{1} \left(\frac{\partial}{\partial x'} \nabla'^2 \chi\right) G(\boldsymbol{r}', \boldsymbol{r}) + \frac{\partial \chi}{\partial x'} \nabla'^2 G(\boldsymbol{r}', \boldsymbol{r}) \Big|_{x'=0} dy'. \tag{7.14}$$

If we now substitute the bi-linear expansion (A1.4) for $G(\boldsymbol{r}', \boldsymbol{r})$ into (7.14) and integrate term by term, we obtain

$$\chi(\boldsymbol{r}) = \sum_{\lambda \in L} A_\lambda E_\lambda(y) e^{-(1/2)\lambda x}, \tag{7.15}$$

where

$$A_\lambda = \frac{\tan^2 \frac{1}{2}\lambda}{2\lambda^3} \int_{-1}^{1} E_\lambda(y) \frac{\partial}{\partial x} \nabla^2 \chi + (2\lambda \cos \tfrac{1}{2}\lambda \cos \tfrac{1}{2}\lambda y) \frac{\partial \chi}{\partial x} \Big|_{x=0} dy, \tag{7.16}$$

the expansion (7.15) certainly being valid for $x > 0$.

LEMMA. *When* $f \equiv 0$, g *is odd, and* $g(\pm 1) = 0$,

$$\frac{\partial}{\partial x} \nabla^2 \chi \Big|_{x=0} = 2g'(y) - 3g'(1) + R(y), \tag{7.17}$$

where $R(y)$ *is a regular function of complex* y *in the domain shown in Figure* 5, *and as* $y \to \pm 1$ *through this domain*

$$R(y) = O(1 - y^2)^{\beta - 2}, \tag{7.18}$$

where $\beta \doteqdot 2.74$.

Proof. Similar to that for the lemma in Theorem 8.

If we now substitute (7.17) into (7.16) we obtain after some integrations by parts,

$$\frac{2\lambda^3}{\tan^2 \frac{1}{2}\lambda} A_\lambda = \int_{-1}^{1} R(y) E_\lambda(y)\, dy - \frac{48}{\lambda^2} \int_{-1}^{1} g''(y) \cos \tfrac{1}{2}\lambda \sin \tfrac{1}{2}\lambda y\, dy$$

$$- \frac{8}{\lambda^2} \int_{-1}^{1} g''(y) E_\lambda'(y)\, dy. \tag{7.19}$$

As $|\lambda| \to \infty$ through $\{\lambda_n\}$, the three integrals on the right in (7.19) are respectively $O(1/n^{\beta-1})$, $O(1/n^2)$, $O(1/n \log n)$ as $n \to \infty$. Hence

$$A_{\lambda_n} = O\left(\frac{1}{n^4 \log n}\right) \tag{7.20}$$

as $n \to \infty$, and this establishes the theorem.

COROLLARY. *If* $g''' \in BV[-1, 1]$ *then*

$$A_{\lambda_n} = O\left(\frac{1}{n^{\beta+2}}\right) \tag{7.21}$$

as $n \to \infty$, *and the terms of* (7.11), (7.12) *are*

$$O\left(\frac{1}{n^{\beta-1} \log n}\right)$$

as $n \to \infty$, *uniformly for* $y \in [-1, 1]$. (*This rate of convergence is not in general improved by placing further smoothness restrictions on* g).

Proof. If $g''' \in BV[-1, 1]$, the last term in (7.19) may be integrated by parts and now is $O(1/n^2 \log n)$. The first term is now the leading contribution.

THEOREM 11. *Suppose* g(y) *is odd with* $g(\pm 1) \neq 0$ *and* $g''(y) \in BV[-1, 1]$. *Then Problem P possesses a solution for the data* $\{0, g\}$. *The data stresses have an eigenfunction expansion of the form* (7.11), (7.12), *where these series are conditionally convergent for each* $y \in (-1, 1)$ *and uniformly convergent for* y *bounded away from* ± 1.

Proof. In view of Theorem 10, it suffices to show that the above result holds for *any particular* g satisfying these conditions. Consider the particular stress field generated by the Airy stress function

$$\hat{\chi}(x, y) = \sum_{\lambda \in L} \frac{4 \tan^2 \frac{1}{2}\lambda}{\lambda^3} E_\lambda(y) e^{-(1/2)\lambda x}, \tag{7.23}$$

which is known (Gregory [9], Theorem 7) to take the end values

$$\hat{\chi}_{xx}(0, y) = 1, \tag{7.24}$$

$$\hat{\chi}_{yy}(0, y) = 0, \tag{7.25}$$

for $-1 < y < 1$. The behaviour of $\hat{\chi}_{xy}(0, y)$ cannot easily be deduced directly from (7.23), but as in previous theorems we may make a comparison with the corresponding right-angled wedge solution. This proves that $\hat{\chi}_{xy}(0, y) \in C^\infty[-1, 1]$ with

$$\hat{\chi}_{xy}(0, 1) = \frac{2}{\pi} \neq 0. \tag{7.26}$$

Thus the theorem certainly holds when $f \equiv 0$, $g = -\hat{\chi}_{xy}(0, y)$, and therefore holds in general.

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Appendix 1. The infinite strip Green's function $G(\mathbf{r}', \mathbf{r})$ and the Papkovich–Fadle eigenfunctions

The infinite strip Green's function $G(\mathbf{r}', \mathbf{r})$ has been investigated thoroughly by the author in [8] and a summary of its properties is given here.

Let $\mathscr{E}$ be the domain $-1<y<1$, $-\infty<x<\infty$. Then

(i) $G(\mathbf{r}', \mathbf{r})$ is infinitely differentiable in $\bar{\mathscr{E}}\times\bar{\mathscr{E}}$ except when $\mathbf{r}'=\mathbf{r}$, and (considered as a function of $\mathbf{r}$, for each $\mathbf{r}'$) satisfies the bi-harmonic equation

$$\nabla^2(\nabla^2 G)=0, \qquad (\mathbf{r}\neq\mathbf{r}'). \tag{A1.1}$$

(ii) For each $\mathbf{r}'\in\mathscr{E}$, $G(\mathbf{r}', \mathbf{r})$ has the form

$$G(\mathbf{r}', \mathbf{r})=R^2\log R+H(\mathbf{r}', \mathbf{r}), \tag{A1.2}$$

where $R=|\mathbf{r}'-\mathbf{r}|$, and H is infinitely differentiable in $\mathscr{E}\times\bar{\mathscr{E}}$.

(iii) For each $\mathbf{r}'\in\mathscr{E}$, G satisfies the boundary conditions

$$G=\frac{\partial G}{\partial y}=0, \qquad (\mathbf{r}\in\partial\mathscr{E}). \tag{A1.3}$$

(iv) For $x\geq x'$, G has the bi-linear expansion

$$G(\mathbf{r}', \mathbf{r})=-\sum_{\lambda\in L}\frac{2\pi\tan^2\frac{1}{2}\lambda}{\lambda^3}\phi_{-\lambda}(\mathbf{r}')\phi_{\lambda}(\mathbf{r}) - \sum_{\mu\in M}\frac{2\pi\cot^2\frac{1}{2}\mu}{\mu^3}\psi_{-\mu}(\mathbf{r}')\psi_{\mu}(\mathbf{r}), \tag{A1.4}$$

where $\{\phi_\lambda(\mathbf{r})\}$, $\{\psi_\mu(\mathbf{r})\}$ are the Papkovich–Fadle eigenfunctions for the infinite strip, given by

$$\phi_\lambda(\mathbf{r})\equiv E_\lambda(y)e^{-(1/2)\lambda x}=\{(y-1)\sin\tfrac{1}{2}\lambda(y+1)+(y+1)\sin\tfrac{1}{2}\lambda(y-1)\}e^{-(1/2)\lambda x} \tag{A1.5}$$

$$\psi_\mu(\mathbf{r})\equiv F_\mu(y)e^{-(1/2)\mu x}=\{(y-1)\sin\tfrac{1}{2}\mu(y+1)-(y+1)\sin\tfrac{1}{2}\mu(y-1)\}e^{-(1/2)\mu x} \tag{A1.6}$$

The sets L, M appearing in (A1.4) are defined by

$$L=\{\lambda;\ \sin\lambda+\lambda=0,\ \mathrm{Re}\,(\lambda)>0\}, \tag{A1.7}$$

$$M=\{\mu;\ \sin\mu-\mu=0,\ \mathrm{Re}\,(\mu)>0\}. \tag{A1.8}$$

Those members of L, M lying in the positive quadrant, and ordered by increasing real part are denoted by $\{\lambda_n\}$, $\{\mu_n\}$ $(n\geq 1)$. Note that $L=\{\lambda_n\}\cup\{\bar{\lambda}_n\}$ and $M=\{\mu_n\}\cup\{\bar{\mu}_n\}$.

Appendix 2. To construct the solution to the boundary value problem for the semi-circular region $\mathscr{D}^C$

The bi-polar transformation (4.7) maps the domain $\mathscr{D}^C$ shown in Figure 3 (1–1) onto the strip $\frac{1}{2}\pi<\eta<\pi$, $-\infty<\xi<\infty$ in the (ξ, η)-plane. The bi-harmonic equation for

the Airy stress function $\chi(x, y)$ is not, of course, preserved by (4.7), but if we introduce the *Jeffery potential* $\phi(\xi, \eta)$ given by

$$\phi = \frac{\chi}{J}, \tag{A2.1}$$

where

$$J(\xi, \eta) = \left|\frac{dz}{d\zeta}\right| = \frac{1}{\cosh \xi - \cos \eta}, \tag{A2.2}$$

then ϕ satisfies

$$\left(\frac{\partial^4}{\partial \xi^4} + 2\frac{\partial^4}{\partial \xi^2 \partial \eta^2} + \frac{\partial^4}{\partial \eta^4} - 2\frac{\partial^2}{\partial \xi^2} + 2\frac{\partial^2}{\partial \eta^2} + 1\right)\phi = 0, \tag{A2.3}$$

an equation with constant coefficients. The resulting boundary value problem in the strip $\frac{1}{2}\pi \leq \eta \leq \pi$ can be solved by taking Fourier transforms in ξ. The formal solution is

$$\begin{aligned}\phi(\xi, \eta) = \frac{\cos \eta}{2\pi} \int_{-\infty}^{\infty} \Phi(\alpha) \Big\{ &-\cosh \alpha(\pi - \eta) \\ &+ \frac{\sinh \tfrac{1}{2}\pi\alpha \cosh \tfrac{1}{2}\pi\alpha \sinh \alpha(\pi - \eta)}{\sinh^2 \tfrac{1}{2}\pi\alpha - \alpha^2} \Big\} e^{-i\alpha\xi}\, d\alpha \\ &+ \frac{\sin \eta}{2\pi} \int_{-\infty}^{\infty} \Phi(\alpha) \frac{\alpha \sinh \tfrac{1}{2}\pi\alpha \cosh \tfrac{1}{2}\pi\alpha}{\sinh^2 \tfrac{1}{2}\pi\alpha - \alpha^2} \\ &\times \{\cosh \alpha(\pi - \eta) - \coth \tfrac{1}{2}\pi\alpha \sinh \alpha(\pi - \eta)\} e^{-i\alpha\xi}\, d\alpha,\end{aligned} \tag{A2.4}$$

where

$$\Phi(\alpha) = \int_{-\infty}^{\infty} \phi(\xi, \pi) e^{i\alpha\xi}\, d\xi, \tag{A2.5}$$

and $\phi(\xi, \pi)$ is known (via (A2.1)) from the data $\{f, 0\}$ in the original boundary value problem in $\mathcal{D}^C$.

From (A2.4) we may deduce all further properties of the solution as required. In particular $\psi(\xi)$, defined by (4.13), is given by

$$\psi(\xi) = -\frac{1}{4\pi} \frac{\partial}{\partial \eta} \left(\frac{\partial^2}{\partial \xi^2} + \frac{\partial^2}{\partial \eta^2}\right)\phi \Big|_{\eta = \pi}, \tag{A2.6}$$

so that $\Psi(\alpha)$, defined by

$$\Psi(\alpha) = \int_{-\infty}^{\infty} \psi(\xi) e^{i\alpha\xi}\, d\xi, \tag{A2.7}$$

is related to $\Phi(\alpha)$ by

$$\Psi(\alpha) = M(\alpha)\Phi(\alpha), \tag{A2.8}$$

where

$$M(\alpha)=\frac{1}{2\pi}\frac{\alpha(\alpha^2+1)\sinh\frac{1}{2}\pi\alpha\cosh\frac{1}{2}\pi\alpha}{\sinh^2\frac{1}{2}\pi\alpha-\alpha^2}. \tag{A2.9}$$

Appendix 3. Analytic continuation of $\int_{-\infty}^{\infty} H(\alpha', \alpha)\Psi(\alpha')\,d\alpha'$

THEOREM 12. *Define*

$$R(\alpha)=\int_{-\infty}^{\infty} H(\alpha', \alpha)\Psi(\alpha')\,d\alpha', \tag{A3.1}$$

where $H(\alpha', \alpha)$ *is given by* (4.25), *and suppose that* $\Psi(\alpha)$ *is analytic in* $S_c (c>0)$ *and that*

$$\Psi(\alpha)=O(e^{-\gamma|\alpha|}) \tag{A3.2}$$

as $|\alpha|\to\infty$ *in* $S_{<c}$ *for each* γ, $0<\gamma<\frac{1}{2}\pi$. *Then* $R(\alpha)$ *is analytic in* S_{c+1}, *except possibly for simple poles at* $\pm i, \pm 2i, \ldots$, *and*

$$R(\alpha)=O(e^{-\gamma|\alpha|}) \tag{A3.3}$$

as $|\alpha|\to\infty$ *in* $S_{<c+1}$ *for each* γ, $0<\gamma<\frac{1}{2}\pi$.

Proof. On applying Parseval's theorem to (A3.1) we obtain

$$R(\alpha)=\int_{-\infty}^{\infty}\psi(\xi')\left\{\int_{-\infty}^{\infty} H(\alpha', \alpha)e^{-i\alpha'\xi'}\,d\alpha'\right\}d\xi', \tag{A3.4}$$

so that

$$\mathcal{V}(\xi)\equiv\frac{1}{2\pi}\int_{-\infty}^{\infty} R(\alpha)e^{-i\alpha\xi}\,d\alpha \tag{A3.5}$$

$$=2\pi\int_{-\infty}^{\infty}\psi(\xi')\frac{\mathcal{G}(\xi', \xi)-\mathcal{G}^C(\xi', \xi)}{J(\xi')J(\xi)}\,d\xi'. \tag{A3.6}$$

Hence $V(y)$, defined by

$$V(\tanh\tfrac{1}{2}\xi)=\mathcal{V}(\xi) \tag{A3.7}$$

is given by

$$V(y)=\frac{4\pi}{1-y^2}\int_{-1}^{1} U(y')\{G(y', y)-G^C(y', y)\}\,dy', \tag{A3.8}$$

where $U(y')$ is given by (4.13). Now it follows from the hypotheses on $\Psi(\alpha)$ that $U(y)$ is analytic in the open lune L_γ (see Figure 4) and

$$U(y)=O((1-y^2)^{c-2-\varepsilon}) \tag{A3.9}$$

as $y\to\pm1$ in L_γ, for arbitrarily small $\varepsilon>0$. Thus since $c>0$, the integral in (A3.8) exists (possibly as a convergent improper integral) for each $y\in L_\gamma$.

Now let $y \to 1$ in L_γ and consider the expression (4.30) for $G - G^C$. The integral term and the middle term make regular contributions to $V(y)$ at $y = 1$, and so it remains to consider the contribution of the log term.

LEMMA. *If* $U(y)$ *satisfies* (A3.9), *then as* $y \to 1$ *in* L_γ

$$\int_{-1}^{1} (y - y')^2 \log\left(\frac{1 - yy'}{2 - y - y'}\right) U(y')\, dy' = \{\text{polynomial in } (1-y)\} + O\{(1-y^2)^{c+2-\varepsilon}\}. \tag{A3.10}$$

This lemma is proved by estimating the above log term when y is near 1. The details are quite lengthy and are omitted.

This lemma now implies that as $y \to 1$ in L_γ,

$$V(y) = (1-y)^{-1}\{\text{polynomial in } (1-y)\} + O\{(1-y^2)^{c+1-\varepsilon}\} \tag{A3.11}$$

but it is evident by considering $\mathscr{V}(\varepsilon)$, (A3.5), that the pole and the constant term in (A3.11) have zero coefficients, and so as $\mathrm{Re}\,(\xi) \to \infty$ in S_γ,

$$\mathscr{V}(\xi) = \mathrm{sech}^2 \tfrac{1}{2}\xi \{\text{polynomial in } \mathrm{sech}^2 \tfrac{1}{2}\xi\} + O\{e^{-|\xi|(c+1-\varepsilon)}\} \tag{A3.12}$$

The theorem now follows by taking the Fourier transform of (A3.12) with respect to ξ.

Appendix 4. The elastic field in a right-angled wedge

Consider the elastic right-angled wedge $0 \le \theta \le \frac{1}{2}\pi$, $0 \le r < \infty$, which is in equilibrium under the surface tractions

$$\sigma_{\theta\theta}(r, 0) = F(r), \tag{A4.1}$$

$$\sigma_{r\theta}(r, 0) = G(r), \tag{A4.2}$$

$$\sigma_{\theta\theta}(r, \tfrac{1}{2}\pi) = \sigma_{r\theta}(r, \tfrac{1}{2}\pi) = 0 \tag{A4.3}$$

$(0 < r < \infty)$, where $F(r)$, $G(r)$ are independent functions prescribed on $\theta = 0$. We will assume that these surface tractions are self-equilibrating and also vanish for all sufficiently large r.

This problem can be solved using the Mellin transform, as in Uflyand [21], Chap. 5. The resulting Airy stress function $\chi(r, \theta)$ is given formally by

$$\chi(r, \theta) = \frac{1}{2\pi i} \int_{-i\infty}^{i\infty} X(p, \theta) r^{-p+1}\, dp, \tag{A4.4}$$

where $X(p, \theta)$ has the form

$$X(p, \theta) = A(p) \cos(p+1)\theta + B(p) \cos(p-1)\theta + C(p) \sin(p+1)\theta + D(p) \sin(p-1)\theta \tag{A4.5}$$

the four coefficients A, B, C, D, being determined from the four boundary conditions (A4.1)–(A4.3).

All further properties of the resulting stress field can be deduced from (A4.4). In particular it follows after some manipulation that:

(i) When $G(r) \equiv 0$,

$$\nabla^2 \chi\Big|_{\theta=0} = 2F(r) + \frac{1}{2\pi i}\int_{-i\infty}^{i\infty} \frac{2p(p+1)\bar{F}(p)}{\sin^2 \frac{1}{2}\pi p - p^2} \frac{dp}{r^{p+1}}, \tag{A4.6}$$

(ii)
$$\frac{1}{r}\frac{\partial}{\partial\theta}\nabla^2\chi\Big|_{\theta=0} = 2G'(r) - \frac{1}{2\pi i}\int_{-i\infty}^{i\infty} \frac{2p(p^2-1)\bar{G}(p)}{\sin^2 \frac{1}{2}\pi p - p^2} \frac{dp}{r^{p+2}}$$
$$- \frac{1}{2\pi i}\int_{-i\infty}^{i\infty} \frac{(p+1)\sin \pi p \bar{F}(p)}{\sin^2 \frac{1}{2}\pi p - p^2} \frac{dp}{r^{p+2}}, \tag{A4.7}$$

where

$$\bar{F}(p) = \int_0^\infty F(r) r^p \, dr, \tag{A4.8}$$

$$\bar{G}(p) = \int_0^\infty G(r) r^p \, dr. \tag{A4.9}$$

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